Supplementary- ADXpertNet: Deep Learning Approach for capturing local and global spatial relationship between FTIR spectra and component data of milk for adulterant detection

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Overview This material provides additional details supporting the experiments presented in this work. **Appendix A** describes the four public datasets used in our analysis. **Appendix B** presents the detailed performance results for additional evaluation metrics, including F1-score, recall, and precision, for both binary and multiclass classification tasks across all baseline models and the proposed model.

1 Appendix A: Details of datasets

Tropical Soil VNIR Dataset : This dataset contains spectral measurements of tropical soil samples acquired using dry-chemistry techniques, along with laboratory-measured soil fertility attributes for 102 samples. It includes spectral data and chemical properties, such as nutrient concentrations and other fertility indicators, enabling analysis of relationships between spectral signatures and soil composition. In our study, the dataset is used to predict the clay content of soil.

The soil dataset used for additional evaluation was obtained from the Mendeley Data repository: https://data.mendeley.com/datasets/88c5kvmgbf/1. We utilized the soil fertility data together with the VNIR spectral data of 102 soil samples. The prediction task involves estimating the clay content using VNIR spectral measurements along with soil component features, such as organic Matter (OM), cation exchange capacity (CEC), pH, base saturation (V), and extractable nutrients such as phosphorus (ex-P), potassium (ex-K), calcium (ex-Ca), and magnesium (ex-Mg). These component variables are used with the VNIR spectra data.

Milk MIR Dataset: This dataset contains individual cow milk records along with associated milk composition and spectral information. The dataset includes metadata such as sampling date and milking time, as well as milk quality parameters including protein content, milk color, and various technological traits. In addition, it is used to predict the color of bovine milk, leveraging the relationship between traditional milk quality traits and spectra.

The milk dataset was obtained from the Government of Ireland open data portal: (https://data.gov.ie/dataset/milk-quality-traits-and-mid-infrared-spectral-data). From this repository, we used the file "Milk_MIR_Traits_Data.csv", which contains milk quality traits with mid-infrared (MIR) spectral measurements. We selected the spectral data within the wavenumber range 941–3822. We utilized Protein content, Total Solids, Fat content, Urea Content, Casein micelle size, Heat stability, pH, and Casein content as component data. These components were combined with the spectral data to predict the color of milk.

Since some component values were missing, we applied a mean imputation to handle the missing entries before performing the regression task. Specifically, missing values in the component data were replaced using the mean of the corresponding feature.

Milk NIR Dataset: The third dataset was obtained from the Zenodo repository: `https://zenodo.org/records/8263430`. We use the spectral variables Trans_tot_1 to Trans_tot_256 as the NIR spectral features. Along with the spectral data, the available milk constituent variables are used as component features. The prediction task in this dataset involves estimating the fat content (FAT) using the combined information from both the spectral measurements and the component features.

This dataset consists of raw milk samples for which laboratory reference measurements were conducted by a milk control center. The spectral information was collected using a Near Infrared Spectroscopy (NIR) analyzer operating in transmittance mode. It includes NIR spectra along with reference measurements of milk composition, enabling the evaluation of predictive models that relate spectral signatures with physicochemical attributes of milk.

These datasets with FTIR data provide diverse spectral and compositional information from different domains, enabling a comprehensive assessment of the proposed model's performance and demonstrating its applicability beyond the primary milk adulteration detection task.

2 Appendix B: Results on other metrics

Since evaluation metrics F1, recall, and precision are not reported in the existing state-of-the-art models, a direct comparison with SOTA methods is not possible. Therefore, we compare the performance of our model with the baseline models as shown below in Table 1 and Table 2.

The results in Table 1 show that the proposed AdXpertNet achieves the best performance across all evaluation metrics compared with both ML and DL baseline models. Among the classical machine learning methods, SVM (Linear) and Logistic Regression perform strongly with F1-scores above 97%, indicating that linear decision boundaries capture a significant portion of the discriminative information in the dataset. However, simpler models such as Naive Bayes and

Table 1. Comparison of models using F1 Score, Recall, and Precision for binary classification on the FTIR dataset. The best results are shown in bold.

Model Type	F1 Score	Recall	Precision
ML Baseline Models on Composite Data			
Naive Bayes	64.78	65.68	66.82
Decision Tree	87.37	87.37	87.54
Logistic Regression	97.05	97.05	97.08
SVM (Linear)	97.60	97.61	97.62
SVM (RBF)	96.37	96.40	96.40
XGBoost	95.68	95.70	95.73
DL Baseline Models on Composite Data			
CNN	97.94	97.96	97.94
CNN + LSTM	98.63	98.63	98.63
CNN + GRU	97.26	97.26	97.26
ADXpertNet	**99.31**	**99.33**	**99.31**

Decision Tree exhibit substantially lower scores, suggesting limited capability in modeling the complex relationships present in the spectral and component data.

Deep learning baselines further improve performance, with CNN + LSTM achieving the best results among the baseline architectures (F1-score of 98.63%), demonstrating the benefit of capturing sequential dependencies in spectral signals. Nevertheless, AdXpertNet outperforms all baseline models, achieving an F1-score of 99.31%, recall of 99.33%, and precision of 99.31%. This improvement indicates that the proposed multimodal architecture effectively leverages both spectral and component information while capturing both local and global feature interactions, resulting in more accurate and reliable adulteration detection.

The results in Table 2 compare the performance of different models for the multi-class classification task. Among the traditional machine learning approaches, SVM (Linear) achieves the best performance with an F1-score of 93.87%, followed by Logistic Regression with 90.12%. In contrast, simpler models such as Naive Bayes and Decision Tree show considerably lower performance, indicating that multi-class adulterant detection is more challenging and requires models capable of capturing complex relationships in the data.

The deep learning baselines further improve the results, with CNN and CNN + LSTM achieving F1-scores of 93.51% and 93.55%, respectively, highlighting the benefit of learning hierarchical and sequential spectral features. However, the proposed AdXpertNet achieves the best overall performance, obtaining an F1-score of 95.61%, recall of 94.94%, and precision of 96.54%. These results demonstrate that the multimodal architecture effectively integrates spectral and component information, enabling the model to better distinguish between mul-

Table 2. Comparison of models using F1 Score, Recall, and Precision for multi-class classification on the FTIR dataset. The best results are shown in bold.

Model Type	F1 Score	Recall	Precision
ML Baseline Models on Composite Data			
Naive Bayes	34.15	45.85	41.90
Decision Tree	70.69	72.76	70.43
Logistic Regression	90.12	91.01	90.54
SVM (Linear)	93.87	94.11	94.56
SVM (RBF)	80.47	80.64	81.78
XGBoost	84.43	84.54	85.63
DL Baseline Models on Composite Data			
CNN	93.51	92.63	94.49
CNN + LSTM	93.55	93.74	93.49
CNN + GRU	90.32	89.98	90.77
ADXpertNet	**95.61**	**94.94**	**96.54**

tiple adulterant classes and achieve more accurate predictions compared to the baseline approaches.

Overall, the results from both binary and multi-class classification demonstrate the effectiveness of the proposed AdXpertNet model for adulteration detection. While high performance is achieved in the binary task due to the simpler distinction between adulterated and unadulterated samples, the model also maintains strong performance in the more challenging multi-class setting where multiple adulterant types must be distinguished. The consistent improvement over baseline ML and DL models in both tasks highlights the capability of AdXpertNet to effectively learn complementary information from spectral and component data, resulting in robust and accurate adulterant detection.